Summary of Lecture 5: The System Bus

1. The System Bus Model

• **Definition:** A shared pathway (collection of wires) connecting the **CPU, Memory, and I/O**.

• **Key Concept (Master vs. Slave):**

◦ **Master:** The device that *initiates* the transfer (e.g., CPU). Only one master can control the bus at a time.

◦ **Slave:** The device that *responds* to the master (e.g., Memory).

• **Exam Note:** The CPU is typically the Master when fetching instructions.

2. Bus Types by Timing (Very Important for MCQs)

How events are coordinated on the bus:

• **A) Synchronous Bus:**

◦ Uses a **Clock (CLK)** signal.

◦ Events happen at fixed time intervals (Rising/Falling edge of the clock).

◦ *Pros:* High throughput, low overhead.

◦ *Cons:* All devices must run at the same speed (less flexible).

• **B) Asynchronous Bus:**

◦ **No Clock.** Uses **Handshaking** protocol.

◦ **Signals:** Uses **STROBE** (Master says "Ready") and **ACK** (Slave says "Received").

◦ *Pros:* Can connect devices with **different speeds**. Best for irregular intervals (like Keyboards).

◦ *Cons:* **Complex and Expensive** hardware (Standard Exam Q).

• **C) Semi-synchronous Bus:**

◦ Uses a Clock but allows variable time.

◦ **Wait States:** If the slave is slow, it sends a signal to make the master "wait" extra clock cycles.

3. Bus Lines (Functional Groups)

Every bus consists of three main groups of lines:

1. **Data Bus:**

- **Function:** Carries the actual data/instructions.
- **Direction: Bi-directional** (Data flows both ways).
- **Width:** Determines **Processor Performance** (e.g., 32-bit vs 64-bit).

2. **Address Bus:**
   - **Function:** Identifies *where* data is going (Memory Location).
   - **Direction: Unidirectional** (CPU Memory).
   - **Width:** Determines **Memory Capacity** ().
     - *Example:* 24 lines = = 16 MB.

3. **Control Bus:**
   - **Function:** Carries commands (Read, Write, Interrupt, Bus Request, Reset).

4. Dedicated vs. Multiplexed

- **Dedicated:** Separate lines for Address and Data. (Faster, but more wires/pins).
- **Multiplexed:** Uses the *same lines* for Address and Data at different times. (Fewer pins, but slower and complex control).
  - *Example:* **PCI Bus** is multiplexed.

5. Real-World Bus Standards (Memorize Numbers)

- **ISA (Industry Standard Architecture):** Old IBM bus. Started as 8-bit, expanded to 16-bit.
- **PCI (Peripheral Component Interconnect):**
  - 32-bit or 64-bit.
  - **Multiplexed** lines.
- **PCI Express (PCIe):**
  - **Hot Pluggable** (Add/remove without restart).
  - High throughput uses **Serial** connection (Lanes: 1x, 4x, 16x).
  - *Exam Note:* **16x** connector is used for **Graphics Cards** (Video).
- **USB (Universal Serial Bus):**
  - **USB 1.1 (Full speed): 12 Mbps**.
  - **USB 2.0 (Hi-speed): 480 Mbps** (Repeated in Exam 2025).
  - Supports up to **127 devices**.

6. Instruction Cycle (Fetch-Execute)

The sequence of the CPU follows:

1. **Fetch:** MAR <- PC (Move address to MAR), then MBR <- Memory (Get instruction).

2. **Decode:** Control Unit figures out what the instruction is.

3. **Operand Fetch:** Get data from memory if needed.

4. **Execute:** ALU performs the operation.

5. **Store:** Write back.

Exam "Cheat Sheet" (Dr. Shahena's Favorites):

1. **Asynchronous Bus:** Best for irregular devices (Keyboard), uses Handshaking, is Complex/Expensive.

2. **Data Bus:** Bi-directional.

3. **USB 2.0 Speed:** 480 Mbps.

4. **PCIe:** Hot pluggable & Graphics use 16x.

5. **Master:** Initiates the transaction.